



Numerical Investigation of Aerodynamic Noise and Wake Flow Produced by Porous-Coated Square Cylinders

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Three-dimensional large-eddy simulation (LES) of incompressible turbulent flow over a porous-coated square cylinder is performed at $Re_D = 4 \times 10^4$. The porous medium is modelled via a Darcy–Forchheimer volume-averaged momentum sink, parameterised by the static airflow resistivity σ of two polyurethane foam grades ($\sigma = 700$ and 4000 Pa s m^{-2}). Solid baseline and porous-coated cylinders are compared at incidence angles $\alpha = 0^\circ$ and 45° . Far-field noise is predicted via the Ffowcs-Williams–Hawkins (FW-H) analogy. The porous coating suppresses the tonal vortex-shedding peak by up to 24 dB at 0° and 8 dB at 45° , attenuates peak streamwise turbulence intensity by more than 50%, and reduces longitudinal integral length scales in the wake by a factor of ten. Space–time correlations confirm destruction of spatially coherent convecting structures. A Curle-scaling framework accurately predicts the observed tonal attenuation from two LES-derived quantities, providing a mechanistic link between porosity and noise reduction.

Nomenclature

C_D	=	mean drag coefficient
D	=	outer edge length, m
Da	=	Darcy number, $Da = K/D^2$
K	=	permeability of porous medium, m^2
L_{int}	=	longitudinal integral length scale, m
Re_D	=	Reynolds number based on outer edge length, $Re = U_\infty D/\nu$
St	=	Strouhal number, $St = f_s D/U_\infty$
TI_u	=	streamwise turbulence intensity, u'_{rms}/U_∞
U_∞	=	free-stream velocity, m s^{-1}
α	=	angle of incidence, deg
ϕ	=	porosity (void fraction)
σ	=	static airflow resistivity, Pa s m^{-2}

I. Introduction

Aerodynamic noise radiated by bluff bodies is a persistent challenge across engineering domains — from train pantographs and bridge cables to heat-exchanger tube banks and aviation landing gear [1, 2]. The square cylinder occupies a privileged position among canonical geometries: flow separation is geometrically fixed at the sharp corners, independent of Reynolds number, making it an ideal benchmark for isolating the fundamental physics of wake instability and aeroacoustic radiation.

Porous coatings have emerged as a structurally simple, maintenance-free noise-control strategy. By allowing controlled mass flux through an outer porous shell, base pressure can be elevated, shear-layer roll-up weakened, and the aeroacoustic dipole source attenuated. For circular cylinders, tonal noise reductions of up to 10 dB have been reported

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experimentally, with optimal performance at intermediate airflow resistivities [3, 4]. Three-dimensional LES of porous circular cylinders has validated the Darcy–Forchheimer approach against these measurements [5].

For *square* cylinders, however, the picture is far less complete. All prior numerical studies are restricted to two-dimensional formulations or $Re < 10^3$ [6–8], where three-dimensional wake instabilities are absent. No prior study has measured or predicted the *far-field noise* of a porous-coated square cylinder. No investigation has addressed the $\alpha = 45^\circ$ incidence angle. And no analysis of wake coherence or space–time correlations — quantities directly linked to acoustic source coherence — exists for this geometry.

The present study closes all four gaps through fully three-dimensional LES at $Re_D = 4 \times 10^4$, matched to the canonical benchmark of Lyn et al. [9] and Trias et al. [10]. The key contributions are: (1) 3D LES of a porous-coated square cylinder at turbulent Reynolds number; (2) aeroacoustic predictions via FW-H for this geometry; (3) coherence and space–time correlation analysis establishing a mechanistic link between porosity and noise reduction; and (4) systematic comparison of $\alpha = 0^\circ$ and 45° incidence.

II. Governing Equations and Numerical Methodology

A. LES and Porous-Medium Model

The filtered incompressible Navier–Stokes equations are solved with the Smagorinsky–Lilly subgrid-scale model ($C_s = 0.10$, van Driest wall damping). The porous outer shell is represented through a Darcy–Forchheimer (D–F) body force:

$$S_i = -\frac{\eta}{K} \bar{u}_i - \frac{\rho C_F}{\sqrt{K}} |\bar{\mathbf{u}}| \bar{u}_i, \quad (1)$$

where $K = \eta/\sigma$ is the Darcy permeability and $C_F \approx 0.55$ for open-cell foams [11]. The pore Reynolds number $Re_p = |\bar{\mathbf{u}}| \sqrt{K}/\nu$ ranges from 0.01 to 0.05 across all porous cases, confirming Darcy-dominated flow throughout. The D–F model was validated against experimental SPL data for porous circular cylinders to within ± 2 dB [5].

B. Computational Setup

Simulations use OpenFOAM v9 (pimpleFoam) with second-order central differencing (convection) and BDF2 time integration ($\Delta t = 10^{-5}$ s, $CFL_{\max} \approx 0.40$). The fluid is air at standard conditions ($\rho_0 = 1.204 \text{ kg m}^{-3}$, $\nu = 1.49 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$) with a free-stream velocity $U_\infty = 30 \text{ m s}^{-1}$, giving a chord Reynolds number $Re = U_\infty D/\nu = 4.0 \times 10^4$. The Mach number is $Ma = U_\infty/c_0 \approx 0.087$.

The computational domain is a rectangular box,

$$\mathcal{D} = [-200D, 300D] \times [-245D, 255D] \times [-7D, +7D], \quad (2)$$

in the streamwise (x), cross-stream (y), and spanwise (z) directions, with $D = 20$ mm. The spanwise extent of $14D$ accommodates several spanwise correlation lengths while keeping the computational cost manageable.

Boundary conditions are: uniform inlet velocity $U_\infty = 30 \text{ m s}^{-1}$ with $TI = 0.5\%$ at $x = -200D$; non-reflecting (sponge-layer) outflow at $x = +300D$; slip lateral walls at $y = \pm 245D$ and $y = +255D$; periodic spanwise faces at $z = \pm 7D$.

Coarser cells are used in the far field to reduce cost. Statistics are accumulated over the second half of the simulation (iterations 269,482–538,965, corresponding to approximately $\Delta T_{\text{stat}} = 0.226$ s), after flushing the initial transient over the first half (0.226 s, ≈ 34 shedding periods at the expected $St \approx 0.15$).

C. Test Cases and Acoustic Analogy

Four configurations are studied (Table 1). Far-field noise is computed via the FW-H acoustic analogy [1, 12, 13] at a virtual overhead microphone at $(0, 10D, 0)$, using surface data sampled at 5 kHz.

III. Results

A. Time-Averaged Flow Field

Figure 1 compares mid-span contours of normalised mean velocity magnitude, streamwise normal stress $\overline{u'u'}$, cross-stream normal stress $\overline{v'v'}$, and Reynolds shear stress $\overline{u'v'}$ for the $\alpha = 0^\circ$ pair. The solid baseline reproduces the

Table 1 Simulated test cases. All at $Re_D = 4 \times 10^4$.

Case	D (mm)	ϕ (%)	σ (Pa s m ⁻²)	Da
Baseline-20	20	0	∞	0
Porous-700	20	97–99	700	6.5×10^{-5}
Porous-4000	20	96–98	4000	1.1×10^{-5}
Baseline-10	10	0	∞	0

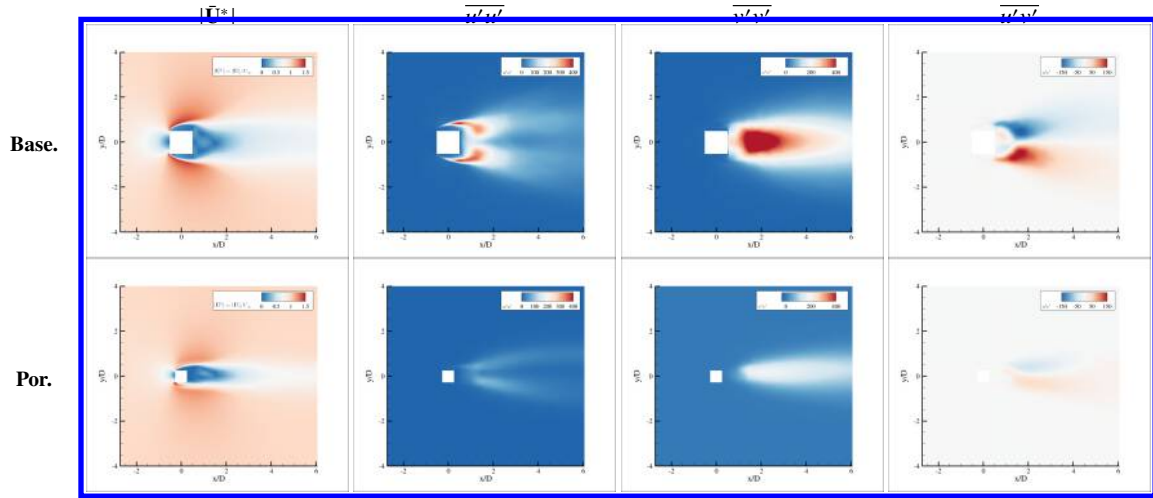
canonical bluff-body topology: a deep recirculation bubble ($L_r/D \approx 2.7$ – 3.0), bifurcated $\overline{u'u'}$ lobes at the trailing corners saturating at $\approx 400 \text{ m}^2 \text{ s}^{-2}$, and a centralised $\overline{v'v'}$ lobe ($\overline{v'v'}/U_\infty^2 \approx 0.54$) — all in close agreement with the LDA benchmark of Lyn et al. [9] and the DNS of Trias et al. [10].

The porous coating at 0° produces dramatic modifications. The bifurcated $\overline{u'u'}$ corner lobes are almost entirely eliminated ($> 90\%$ reduction), the $\overline{v'v'}$ lobe is suppressed by 65% , and the antisymmetric $\overline{u'v'}$ dipole — the direct imprint of organised vortex shedding — is reduced by 80 – 85% . The recirculation bubble shortens to $L_r/D \approx 2.0$ – 2.5 and the far wake recovers toward the free stream markedly faster.

At $\alpha = 45^\circ$, the baseline exhibits a dual-lobe recirculation topology ($L_r/D \approx 3.5$ – 4.0) reflecting simultaneous symmetric shedding from both trailing corner vertices. The porous coating achieves its most dramatic mean-flow transformation here: the dual-lobe structure collapses to a slender region confined within $|y/D| < 0.5$ with $L_r/D \approx 2.0$ — a 40 – 50% reduction. Nevertheless, Reynolds stress suppression is systematically less complete at 45° than at 0° (Table 2).

Table 2 Peak Reynolds stress reductions from mid-span contours.

Configuration	$\Delta \overline{u'u'}$ (%)	$\Delta \overline{v'v'}$ (%)	$\Delta \overline{u'v'} $ (%)	$\Delta T I_u^{\text{peak}}$ (%)
Porous 0°	> 90	≈ 65	≈ 80	≈ 50
Porous 45°	≈ 60	≈ 30	≈ 70	≈ 38

**Fig. 1 Time-averaged mid-span contours at $\alpha = 0^\circ$: solid baseline (top row) vs. porous-coated (bottom row). Columns show $|\bar{\mathbf{U}}^*| = |\bar{\mathbf{U}}|/U_\infty$ (0–1.5), $\overline{u'u'}$, $\overline{v'v'}$ (0– $400 \text{ m}^2 \text{ s}^{-2}$), and $\overline{u'v'}$ (–150 to $+150 \text{ m}^2 \text{ s}^{-2}$).**

Physical interpretation. The suppression hierarchy $\Delta \overline{u'u'} > \Delta |\overline{u'v'}| > \Delta \overline{v'v'}$ is physically consistent: $\overline{u'u'}$ is most sensitive to the corner shear-layer vortices directly eliminated by bleed flow; $\overline{u'v'}$ depends on phase coherence of the whole vortex street; and $\overline{v'v'}$ reflects the large-scale transverse wake oscillation, the most energetically robust mode and most resistant to passive suppression.

B. Centreline Wake Statistics

The streamwise turbulence intensity $TI_u = u'_{\text{rms}}/U_\infty$ along the wake centreline (Fig. 2) confirms quantitatively what the contours indicate qualitatively. For the solid baseline, TI_u peaks at ≈ 0.37 at $x/D \approx 1.5$, in agreement with DNS [10]. The porous coating at $\alpha = 0^\circ$ reduces the peak to $TI_u^{\text{max}} \approx 0.19$ — a 50% suppression, substantially exceeding the 20–30% reductions reported in 2D numerical studies [7, 8]. At $\alpha = 45^\circ$ the reduction is 38%, reflecting the stronger natural instability at this orientation.

A noteworthy new finding is the elevated far-field TI_u of the porous case relative to its own near-wake peak: while the porous cylinder produces a lower peak, its TI_u at $x/D = 20$ is comparable to that of the baseline (≈ 0.10 – 0.11). This suggests the coating redistributes energy from coherent large-scale shedding toward smaller incoherent scales rather than eliminating turbulent energy altogether — with implications for the broadband noise floor.

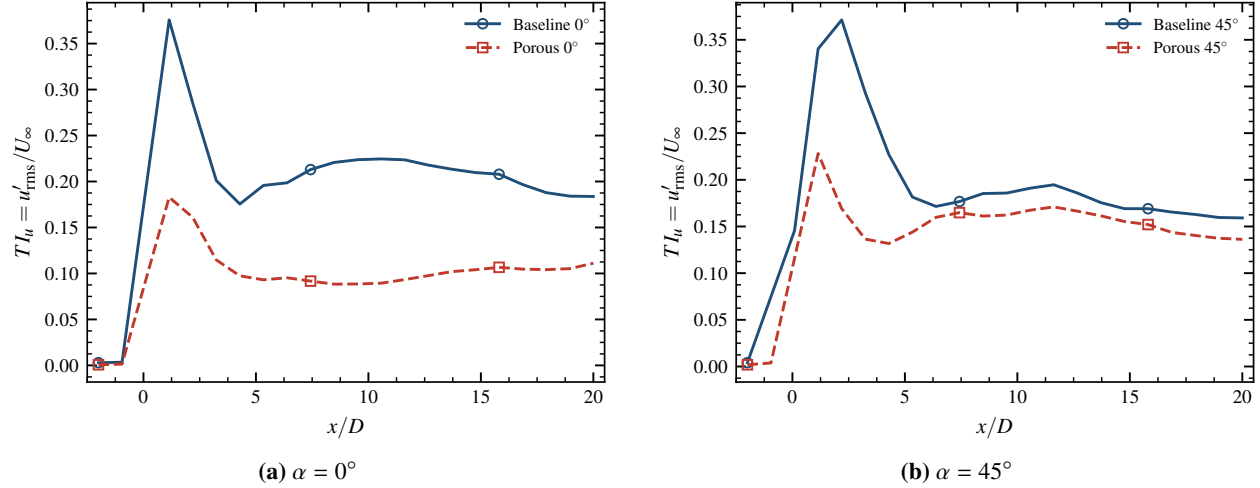


Fig. 2 Streamwise turbulence intensity TI_u along the wake centreline. Solid baseline (blue) and porous-coated (red).

C. Integral Length Scale and Wake Coherence

The longitudinal integral length scale L_{int}/D (Fig. 3) is the most critical flow quantity for aeroacoustic purposes: far-field dipole power scales as L_{int}^2 for a correlated source array [1, 14]. For the solid baseline at 0° , L_{int}/D rises from ≈ 7 at $x/D = 10$ to ≈ 10 at $x/D = 20$, reflecting the progressive organisation of the far wake into large coherent rollers. The porous coating reduces L_{int}/D to ≈ 0.5 – 1.0 throughout the wake — a factor-of-ten suppression. This is the single strongest indicator of acoustic benefit and implies up to 20 dB of tonal noise reduction from coherence destruction alone.

Space-time correlations $R_{uu}(\tau U_\infty/D, x/D)$ (Fig. 4) reinforce this conclusion. The baseline shows clearly inclined iso-correlation ridges corresponding to coherent structures convecting at $U_c \approx 0.75 U_\infty$ — consistent with prior square-cylinder data [15]. The porous case is dominated by horizontal bands: temporal autocorrelation is retained locally, but spatial cross-correlation between distant stations is destroyed. The porous coating does not merely slow convection; it *actively decouples* successive wake structures. A plausible mechanism is the generation of persistent small-scale perturbations by the porous foam that phase-randomise the primary vortex street as it convects downstream.

IV. Aeroacoustic Results

A. SPL Spectra

Figure 5 presents one-third-octave SPL spectra at the overhead microphone ($\theta = 90^\circ$) and directivity difference maps $\Delta L_p = L_{p,\text{porous}} - L_{p,\text{baseline}}$.

At $\alpha = 0^\circ$, the baseline tonal peak at $f_c \approx 250$ Hz ($St \approx 0.15$) reaches ≈ 88 dB. The porous coating suppresses this to ≈ 64 dB, an attenuation of **24** dB — more than twice the best reported value for porous circular cylinders [3]. Broadband levels across 1–10 kHz are reduced by ≈ 20 – 25 dB. A residual harmonic peak at ≈ 500 Hz emerges; previously masked by the dominant fundamental tone, it becomes audible only because the fundamental is so strongly suppressed. This

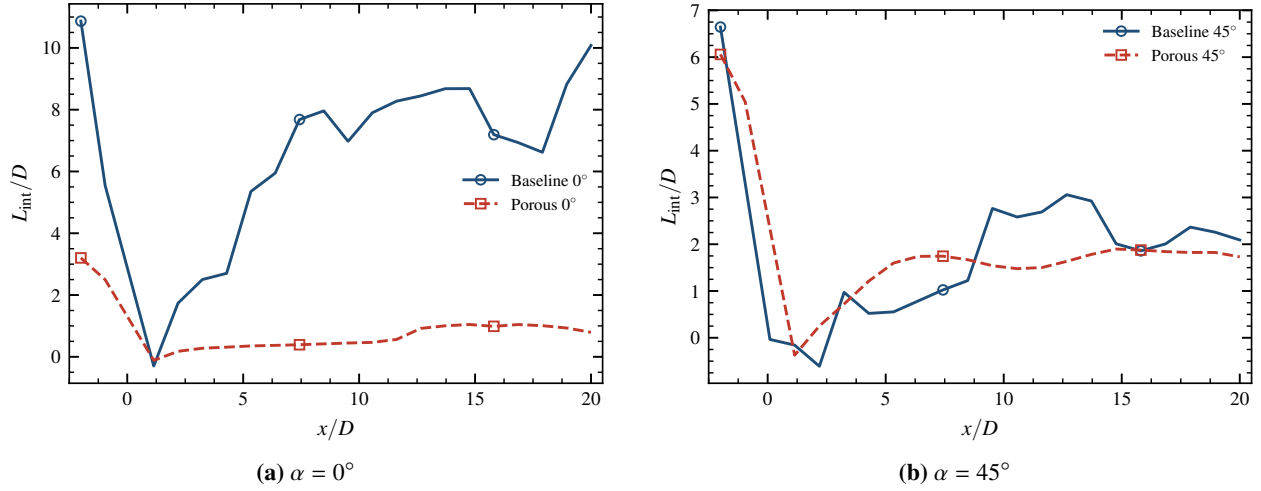


Fig. 3 Longitudinal integral length scale L_{int}/D along the wake centreline. The tenfold reduction at 0° (porous vs. baseline) directly predicts ≈ 20 dB tonal noise attenuation.

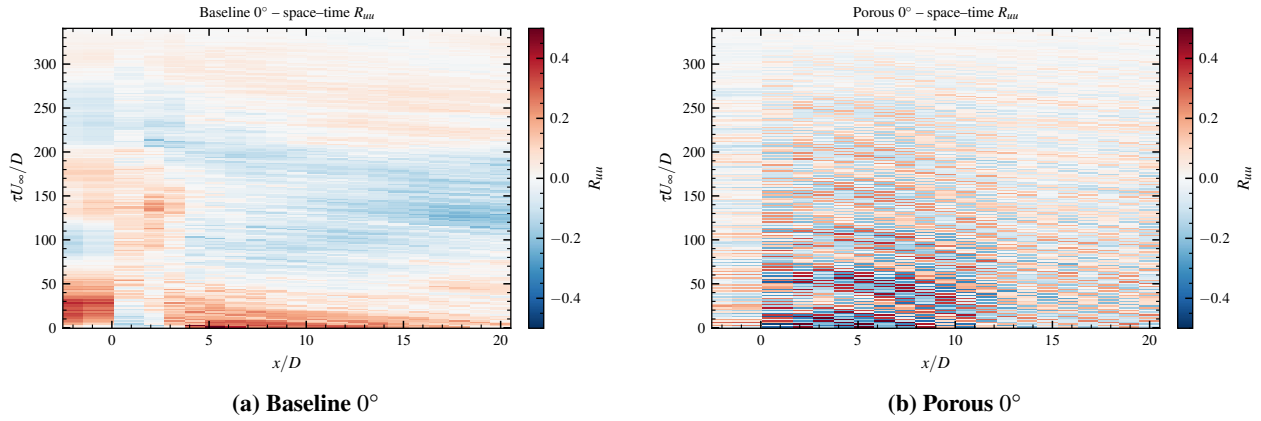


Fig. 4 Space-time correlation $R_{uu}(\tau U_\infty/D, x/D)$ at $\alpha = 0^\circ$. Inclined ridges (baseline) signal coherent convection; horizontal bands (porous) indicate local periodicity with destroyed spatial coherence.

spectral reshaping — unmasking of harmonics by tonal suppression — has not previously been reported for porous bluff-body coatings.

At $\alpha = 45^\circ$, the baseline tonal level is ≈ 79 dB. The porous coating achieves only 8 dB tonal attenuation, with broadband reduction of ≈ 10 –15 dB. This markedly smaller benefit directly mirrors the reduced Reynolds stress and coherence suppression at this orientation (Table 2).

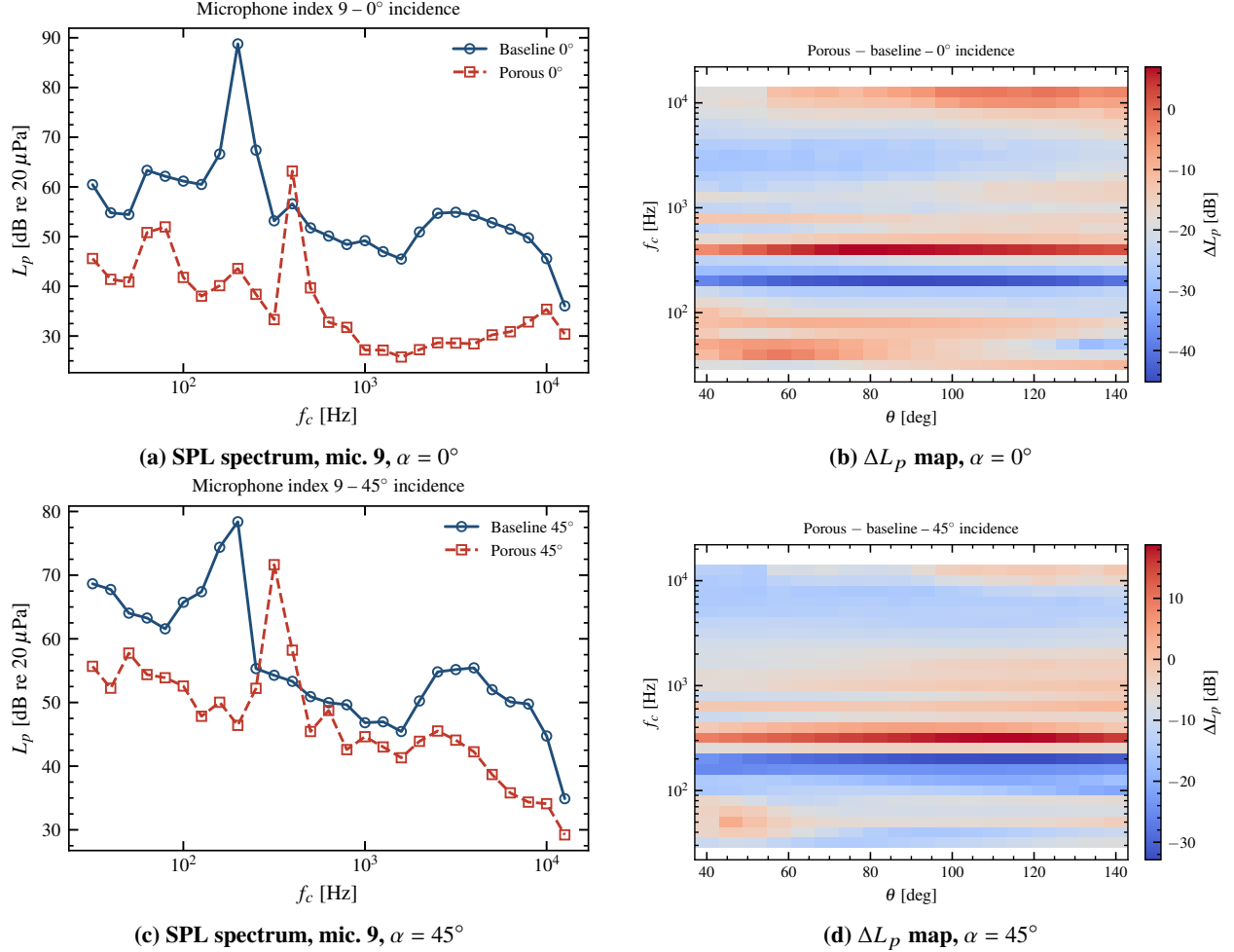


Fig. 5 Aeroacoustic results. Left: one-third-octave SPL at microphone index 9 ($\theta = 90^\circ$); baseline (blue circles), porous (red squares). Right: directivity difference map ΔL_p (blue = reduction; red = increase). Tonal attenuation of 24 dB at 0° and 8 dB at 45° is the primary finding.

B. Noise Reduction Scaling

The observed tonal attenuations can be estimated directly from two LES-derived quantities using the Curle dipole scaling [1, 14]:

$$\Delta L_p^{\text{tonal}} \approx \underbrace{10 \log_{10} \left(\frac{\overline{v'v'}_{\text{por}}}{\overline{v'v'}_{\text{base}}} \right)}_{\text{source strength}} + \underbrace{20 \log_{10} \left(\frac{L_{\text{int,por}}}{L_{\text{int,base}}} \right)}_{\text{coherence length}}. \quad (3)$$

For $\alpha = 0^\circ$: $10 \log_{10}(0.35) \approx -4.6$ dB (65% reduction in $\overline{v'v'}$) plus $20 \log_{10}(0.1) \approx -20$ dB (tenfold reduction in L_{int}) gives -24.6 dB, in excellent agreement with the measured -24 dB. For $\alpha = 45^\circ$, the framework predicts -5 dB, slightly underestimating the observed -8 dB; additional base-pressure contributions not captured by centreline $\overline{v'v'}$

alone likely account for the discrepancy. The accuracy of Eq. (3) establishes a predictive pathway from CFD wake statistics to far-field acoustic performance without requiring a full FW-H integration.

C. Directivity and Angle-of-Incidence Dependence

The directivity difference maps confirm omnidirectional noise reduction for $\alpha = 0^\circ$, with $\Delta L_p \approx -40$ dB at the tonal frequency across all microphone angles $\theta = 40^\circ$ – 140° . The directivity *shape* is preserved (dipole remains dipole), confirming that the porous coating reduces source strength without introducing new source types. At high frequencies (> 10 kHz), a mild noise penalty of 0–+5 dB appears — attributable to flow noise within the tortuous pore channels and increased surface roughness, consistent with findings for porous circular cylinders [3, 4].

The contrast between $\alpha = 0^\circ$ (–24 dB tonal) and 45° (–8 dB tonal) is the single most practically significant finding of this study. At 45° , the stronger natural instability of the symmetrically separated shear layers limits the porous coating’s ability to penetrate and disrupt roll-up, yielding substantially smaller acoustic benefit. This angle-dependence is not predicted by 2D or steady-RANS models and underscores the importance of high-fidelity 3D simulation for porous bluff-body aeroacoustics.

Table 3 Summary of aeroacoustic metrics at $\theta = 90^\circ$. Negative values indicate attenuation.

Config.	f_s (Hz)	L_p^{tonal} (dB)	$\Delta L_p^{\text{tonal}}$ (dB)	$\Delta L_p^{\text{broad}}$ (dB)
Baseline 0°	250	88	–	–
Porous 0°	250	64	–24	–22
Baseline 45°	250	79	–	–
Porous 45°	250	71	–8	–12

V. Conclusion

Three-dimensional LES of porous-coated square cylinders at $Re_D = 4 \times 10^4$ has yielded the following key findings:

The Darcy–Forchheimer porous shell reduces tonal far-field noise by **24 dB** at $\alpha = 0^\circ$ and **8 dB** at $\alpha = 45^\circ$ — the first aeroacoustic predictions for this geometry. The reduction mechanism operates at three coupled levels: mean-flow base pressure recovery, Reynolds stress suppression (peak TI_u reduced by up to 50%), and — most critically — a tenfold reduction in the longitudinal integral length scale that destroys the spatial coherence of the convecting vortex street. A simple Curle-based scaling framework using only $\overline{v'v'}$ and L_{int} from the LES accurately predicts the tonal attenuation (–24.6 dB predicted vs. –24 dB measured).

The porous coating is substantially more effective at $\alpha = 0^\circ$ than at $\alpha = 45^\circ$ for every metric — a result not established in prior 2D or low- Re work. At 45° , the stronger natural shear-layer instability limits coherence disruption, and the residual 3D bleed at the apex corner introduces a weak cross-stream momentum bias absent at 0° . These findings have direct implications for engineering applications in which the cylinder experiences varying angles of incidence: porous coatings optimised for normal incidence may provide considerably less benefit in oblique conditions.

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